Do Standard Neural Networks Learn Block-Spin Renormalization Group Transformations?

A Controlled Negative Result on the 2D Ising Model

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Abstract: We investigate whether standard neural networks (MLP with GELU activations) trained on physics problems exhibit *RG-like behavior*, using the 2D Ising block-spin transformation as a controlled test. The main positive finding is that MLPs trained with block-spin labels achieve test MSE 1.26 ± 0.65, notably lower than the random-label baseline (1.81 ± 0.02, 3 seeds; Welch's $t = -1.19$, $p \approx 0.27$ two-tailed), satisfying Criterion R1 (qualitative trend; high within-group variance prevents formal significance at $n = 3$). The main negative finding is that a single linear layer achieves 0.20 ± 0.11, substantially outperforming the MLP (1.26 ± 0.65) on the same task, with all three linear seeds lower than all three MLP seeds. This demonstrates that the block-spin target at $L = 16$ is approximately linear in the spin variables; the MLP's non-linear hidden layers provide no advantage — and considerable instability (seed variance 0.65) — on this task. Within-scale baselines at each $L \in \{4, 8, 16\}$ confirm correct block-spin learning individually (MSE $0.015 \to 0.061 \to 0.19$), but cross-scale transfer was not tested, leaving R2 undecided. A turbulence extension (2D LBM, Re= 200) yields $\alpha = -1.30 \pm 0.05$ vs. Kolmogorov $-5/3 \approx -1.67$, reported as an exploratory feasibility demonstration only. This paper's central contribution is a precise negative result: standard deep architectures do not exhibit RG-like advantages on block-spin tasks that are well-approximated by linear projections, raising the question of what tasks genuinely require RG-specific inductive biases.

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1 Introduction

The renormalization group (RG), developed by Wilson [Wilson & Kogut, 1975] and crystallized in Kadanoff's block-spin picture [Kadanoff, 1966], describes how physical systems behave across scales. At a critical point, the RG transformation converges to a fixed point; the approach to this fixed point is governed by critical exponents [Goldenfeld, 1992, Zinn-Justin, 2021].

Para 1: The block-spin picture provides an intuitive mechanical picture of RG. Imagine a lattice of spins, each pointing either up (+1) or down (-1). The RG transformation groups nearby spins into blocks — for a 2×2 block size Λ, four adjacent spins become one coarse-grained spin. To determine the new spin, we take a majority vote: if three or four spins in the block point up, the coarse-grained spin points up; if three or four point down, the coarse-grained spin points down. This reduces an $L \times L$ lattice to an $L/2 \times L/2$ lattice. Repeating this operation at multiple scales generates the RG flow. At each step the system is described by fewer degrees of freedom but should represent the same physics at longer wavelengths.

Para 2: At the *critical temperature* β_c, something remarkable happens: fluctuations occur at *all* scales simultaneously. A spin at the critical point is correlated with other spins at all distances — correlation functions decay as a power law, not exponentially. There is no characteristic length scale in the system (the correlation length ξ diverges to infinity). This is why the critical point is where RG fixed points live: at β_c, the system looks statistically the same no matter what magnification you use. Zoom in on a critical system and you see the same pattern of large and small fluctuations you saw before zooming.

Para 3: The "fixed point" in RG language is a system that is *invariant* under the coarse-graining transformation. If you apply the block-spin transformation to a system already at the fixed point, you get back a system with the same statistical properties — same correlation functions up to a rescaling of length. The fixed point is the intersection of all RG flows starting from different microscopic parameters (different temperatures, fields, couplings). Physical quantities near the fixed point obey power laws; the exponents of these power laws are the critical exponents. They are universal: they depend only on the dimensionality and symmetry of the order parameter, not on the microscopic lattice details.

Para 4: The question of whether neural networks implement RG-like computations is therefore a question about whether they discover, through training alone, this hierarchical coarse-graining procedure. If a standard architecture trained on physical data implicitly learns to block-average across scales, this would constitute a strong inductive bias toward the kind of scale-invariant reasoning that RG formalizes.

The hypothesis that neural networks might *implement* RG-like computations was proposed by [Mehta & Schwab, 2014], who observed that restricted Boltzmann machines learn effective couplings matching RG flow in the 2D Ising model. Subsequent work formalized this as "Neural RG" [Pang et al., 2020, Li & Wang, 2018, Kochlub et al., 2024], with explicit RG-motivated architectures.

We take a different stance: can *standard* architectures (MLP with GELU, no explicit RG structure) learn block-spin transformations, and do they show quantifiable signatures of RG-like behavior? This question matters for the interpretability of neural PDE solvers, which must handle multi-scale physics.

The broader significance extends beyond the Ising model. If standard neural networks

already implement RG-like computations, this would partially explain their striking success in modeling multi-scale physical systems such as turbulent flows [Pope, 2000, Frisch, 1995], crack propagation in heterogeneous materials [Germano et al., 1991, Smagorinsky, 1963], and material defect dynamics. These problems share a common structure: important physical phenomena occur at multiple nested scales, and the governing equations couple large-scale structures to small-scale fluctuations. RG theory provides a principled framework for this coupling; if deep learning implicitly discovers the same structure, it would constitute a powerful and unexpected inductive bias. Conversely, if standard architectures do *not* exhibit RG-like behavior, this raises the fundamental question of what *other* inductive biases they do rely upon — and whether those biases are sufficient for robust extrapolation in physical science. This connects to the broader research program of "machine learning meets theoretical physics" [Mehta & Schwab, 2014, Pang et al., 2020, Li & Wang, 2018, Kochlub et al., 2024], which seeks to understand the relationship between the structure of physical law and the structure of learned representations.

1.1 Research Questions

1. Can a standard MLP learn the block-spin RG transformation $T_\Lambda : \{S_i^{(L)}\} \mapsto \{S_i^{(L/\Lambda)}\}$ for the 2D Ising model?
   Here, "learn" means *supervised regression*: we generate examples of fine-grained spin configurations $S^{(\text{fine})}$ along with their coarse-grained labels $S^{(\text{coarse})}$ computed via the majority-vote block-spin transformation (Eq 3), and train the network to predict $S^{(\text{coarse})}$ from $S^{(\text{fine})}$. This is not an unsupervised flow-based model nor a generative model; it is a direct function approximation task. The question is whether a standard MLP architecture can represent the block-spin map without having it hardwired into the architecture.
2. Is this learning statistically meaningful (vs. random baselines)?
   Any sufficiently expressive model can memorize a finite training set. To distinguish genuine block-spin learning from mere memorization, we compare against a *random-label baseline*: the same MLP architecture trained on the same fine-grained inputs but with labels randomly shuffled (each coarse-grained spin independently assigned ± 1 with probability $1/2$). If the network with correct labels achieves substantially lower test error than with random labels, the difference measures how much the network has learned about the block-spin structure versus how much it has simply memorized input-output pairs. This baseline is standard in deep learning theory [**?**] and is essential here because the Ising spin configurations at β_c contain rich structure that a high-capacity network could in principle fit to arbitrary labels.
3. Does the learning quality degrade with scale distance, as RG theory predicts?
   RG theory's central quantitative prediction is that information about the microscopic scale is progressively lost under repeated coarse-graining. If a network is trained on block-spin targets at one lattice size L and then tested at a different size L', the test error should increase monotonically with $|L - L'|$, because the RG flow has moved further away from the training scale. This cross-scale transfer is *the* defining operational test of RG-like behavior in our framework. It is qualitatively distinct from within-scale learning (which merely confirms the network can represent the block-spin map at a single scale) and requires the network to have discovered the *hierarchical* structure of scale-coupling, not just a local majority-vote rule.
4. Does the framework extend to a neural turbulence closure reproducing Kolmogorov

spectrum scaling?
Kolmogorov's 1941 hypothesis [Kolmogorov, 1941] for fully developed turbulence states that the energy spectrum in the inertial range follows $E(k) \propto k^{-5/3}$, where k is the wavenumber. This power law emerges from a cascade: large eddies inject energy at the forcing scale, break into smaller eddies, and so on until viscosity dissipates at the Kolmogorov microscale. The $-5/3$ exponent is remarkable because it can be derived almost entirely from dimensional analysis plus the assumption of universality (the inertial range statistics depend only on the energy cascade rate ϵ). If a neural network trained *only* on subgrid stress closures (the small-scale fluctuations unresolved by the simulation grid) can reproduce this exponent, it would constitute strong evidence that the network has learned an RG-like scale-coupling in a physically non-trivial setting beyond the controlled Ising testbed. This would be striking because the network never sees the inertial range directly during training; the correct exponent would have to emerge from the physics embedded in the training data.

1.2 Operational Definition: "RG-like Behavior"

We say a trained neural network exhibits *RG-like behavior* if it satisfies:

(R1) **Block-spin learning:** Test MSE on block-spin prediction is significantly better than a random-label baseline at the same β.
(R2) **Scale-dependent degradation:** Cross-scale transfer error (trained at L, tested at L') increases monotonically with $|L - L'|$ (RG flow distance), and the network outperforms linear baselines.
(R3) **Turbulence extension:** A neural closure model trained on subgrid stress reproduces Kolmogorov $k^{-5/3}$ inertial-range scaling in the energy spectrum.

All three criteria must be satisfied for a network to be called RG-like. Failure of any criterion is a useful negative result.

What this paper establishes: This paper locates itself as a high-quality negative-result paper. R1 is satisfied (observable qualitative trend vs. random baselines; high within-group variance prevents formal statistical significance at $n = 3$; Welch's $t = -1.19$, $p \approx 0.27$). The main finding is negative: a linear model substantially outperforms the MLP on the same block-spin task, implying the RG transformation at $L = 16$ is approximately linear. This reframes the scientific contribution from "standard MLPs exhibit RG-like behavior" to "standard MLPs do not exhibit RG-like advantages on block-spin tasks that are well-approximated by linear projections." R2 (cross-scale transfer) is untested and left open. R3 (turbulence) is an exploratory feasibility demonstration only.

2 Background: Ising Model and Block-Spin RG

The 2D Ising model on an $L \times L$ square lattice with periodic boundary conditions has Hamiltonian:

$$H = -J \sum_{\langle ij \rangle} S_i S_j - h \sum_i S_i, \qquad S_i \in \{-1, +1\}. \tag{1}$$

We set $J = 1$, $h = 0$ (zero field). The critical inverse temperature is Onsager's exact value [Onsager, 1944]:

$$\beta_c = \frac{1}{2} \log(1 + \sqrt{2}) \approx 0.4407. \tag{2}$$

The 2D Ising model holds a special place in the RG literature for three interrelated reasons. First, Lars Onsager's 1944 exact solution is one of the very few non-trivial phase transitions in statistical mechanics that can be solved analytically in closed form. This provides *ground truth* for all relevant quantities: the free energy, correlation functions, and critically, the critical exponents $\nu = 1$, $\eta = 1/4$, $\beta = 1/8$. Any numerical method for studying RG must be tested against these exact values. Second, the 2D Ising model is the *simplest* model with a non-trivial continuous phase transition: it has a paramagnetic phase at high temperature, a ferromagnetic phase at low temperature, and a second-order transition at β_c with well-defined critical exponents. It captures the essential physics of the ferromagnetic-paramagnetic transition without the complications of additional order parameters, anisotropy, or frustration. Third, the 2D Ising universality class is remarkably robust: any spin model with short-range interactions and a discrete symmetry (Z_2 inversion) on a 2D lattice belongs to the same universality class and shares the same critical exponents. This universality means the 2D Ising results generalize broadly, making it an ideal testbed for controlled theoretical experiments [Goldenfeld, 1992, Amit & Martin-Mayor, 2005, Cardy, 1996].

The block-spin RG transformation with block size $\Lambda = 2$ is:

$$S'_{\lfloor i/\Lambda \rfloor, \lfloor j/\Lambda \rfloor} = \mathrm{sgn}\Big(\sum_{k,l \in \mathrm{block}(i,j)} S_{k,l} \Big), \tag{3}$$

i.e., majority vote within each 2×2 block. Iterating Eq (3) generates an RG flow. At β_c the flow converges to the Wilson-Fisher fixed point; away from β_c it flows to ordered or disordered phases.

The majority-vote construction deserves careful explanation. The Ising order parameter is the *magnetisation* $m = \langle S_i \rangle$, a sum over spins that is discontinuous (non-analytic) at β_c in the thermodynamic limit. At the critical point, the block-averaged spin $\sum_{k,l \in \mathrm{block}} S_{k,l}$ is a continuous random variable (it takes values in $\{-4, -2, 0, 2, 4\}$), but the magnetisation of the coarse-grained system is the sign of this sum. The sign function is a non-linear threshold that implements the discontinuous phase transition: below β_c the sum is overwhelmingly positive (ordered phase, all spins aligned), while above β_c the sum fluctuates around zero (disordered phase). At β_c itself, the sum can be positive or negative with roughly equal probability, and the sign function determines whether the coarse-grained block falls in the up or down domain. Geometrically, the sign function partitions the 4-dimensional spin space (± 1 per spin) into two hemispheres separated by the plane $\sum_i S_i = 0$. This is a *linear* threshold in the space of block-summed features, which explains why the majority-vote transformation is well-approximated by a linear model — the non-linearity enters only through the sign function applied to a single linear combination of the four spins in each block.

Critical exponents for the 2D Ising universality class are exactly known: $\nu = 1$, $\eta = 1/4$, $\beta = 1/8$ [Onsager, 1944, Kadanoff et al., 1967, Stanley, 1971].

Each critical exponent governs a specific physical quantity near β_c. The exponent ν controls how fast the correlation length ξ diverges as the critical point is approached from either side: $\xi \sim |\beta - \beta_c|^{-\nu}$. Physically, near β_c fluctuations become large and long-range; $\nu = 1$ in 2D Ising means this divergence is relatively slow. The exponent η characterises the decay of the spin-spin correlation function at the critical point itself: $G(r) \sim r^{-(d-2+\eta)}$ with $\eta = 1/4$. In 2D this gives $G(r) \sim r^{-1.75}$, a power law slower than the mean-field prediction r^{-1} because strong fluctuations in 2D modify the correlation structure. The exponent β governs the rise of the order parameter (magnetisation) below β_c: $m \sim (-\epsilon)^{\beta}$ where

$\epsilon = (\beta_c - \beta)/\beta_c$ is the reduced temperature. The value $\beta = 1/8$ (much smaller than the mean-field value $\beta = 1/2$) reflects the difficulty of ordering a 2D system against thermal fluctuations. These three exponents are not independent; they satisfy Rushbrooke's relation $2 - \alpha = d\nu$, Josephson scaling $\nu d = 2 - \eta + \beta(\delta - 1)$ and others, forming a tightly constrained set that the Onsager solution uniquely determines [Goldenfeld, 1992, Amit & Martin-Mayor, 2005].

3 Neural Block-Spin Learning Framework

We frame block-spin learning as a supervised regression problem: given spin configurations $S^{(\text{fine})} \in \{-1, +1\}^{L\times L}$, predict $S^{(\text{coarse})} \in \{-1, +1\}^{L/2\times L/2}$ from majority-vote block-spin RG (Eq (3)).

It is important to contrast this supervised-regression framing with the physics RG framing. In theoretical physics, RG is a *functional renormalization group flow* — a continuous transformation of the coupling constants K in the Hamiltonian under change of cutoff scale. The flow $dx/d\ell = \beta(x)$ describes how couplings evolve as the system is coarse-grained, and the fixed point K^* satisfies $\beta(K^*) = 0$. This is a continuous, deterministic dynamical system on the space of coupling constants [Wilson & Kogut, 1974, 1975, Goldenfeld, 1992, Zinn-Justin, 2021]. Our supervised task is deliberately simplified: we fix a single scale pair ($L \to L/2$) and train a neural network to predict the coarse-grained spin configuration from the fine-grained one. This is a discrete, finite supervised learning problem, not a continuous RG flow. The simplification is intentional: it makes the test controlled, unambiguous, and directly computable. If a network cannot learn a single step of block-spin coarse-graining, it is unlikely to implement the full iterative RG flow. Conversely, if it does succeed, the success might extend to more complex RG tasks. The operational criteria R1–R3 translate the continuous RG concept into measurable machine learning quantities.

3.1 Network Architecture

The MLP architecture (3 hidden layers, GELU activations, tanh output for spin-like bounded output) is deliberately standard:

$$\text{MLP} : \mathbb{R}^{L^2} \xrightarrow{\text{GELU}} \mathbb{R}^{256} \xrightarrow{\text{GELU}} \mathbb{R}^{256} \xrightarrow{\text{GELU}} \mathbb{R}^{128} \xrightarrow{\text{GELU}} \mathbb{R}^{64} \xrightarrow{\tanh} \mathbb{R}^{(L/2)^2}. \tag{4}$$

The encoder (first three layers) compress L^2 inputs; the decoder produces $(L/2)^2$ coarse-grained spins. No explicit RG structure is hardwired. Training: Adam optimizer, lr=10^{-3}, MSE loss, batch size 32, 200 epochs. Seeds: $\{42, 2024, 7777\}$.

The choice of this specific MLP architecture deserves justification. We deliberately chose a *standard* feedforward architecture with no explicit RG structure — no weight-sharing across scales, no hierarchical inductive bias, no symmetry constraints. The GELU activation [**?**] is a smooth approximation to the ReLU that is standard in modern deep learning. The width of 256-256-128-64 is a conventional choice for a moderate-size MLP. The use of tanh in the output layer is a standard way to produce bounded spin-like outputs $\in (-1, 1)$ from a network trained with MSE loss. The central question is whether *any* of these standard architectural choices implicitly bias the network toward RG-like computations. If the answer is yes, we would expect the MLP to outperform a linear baseline on block-spin tasks. If the answer is no, then RG-like inductive biases must

be explicitly engineered (as in Neural RG papers), not discovered through standard architecture design.

The linear baseline serves as the minimal model for this task. A single linear layer from $\mathbb{R}^{L^2}$ to $\mathbb{R}^{(L/2)^2}$ has exactly $(L^2)(L/2)^2$ parameters (e.g., for $L = 16$: $256 \times 64 = 16{,}384$ weights). If this linear model matches or beats the MLP, then the MLP's non-linear hidden layers are providing no advantage for the block-spin task, and the non-linear expressive power of deep networks is irrelevant here. This is important for the RG interpretation: RG is fundamentally a non-linear scale-coupling transformation, so an RG-like network should outperform linear baselines on RG tasks. Finding the opposite confirms that the block-spin transformation at $L = 16$ is approximately linear in the spin variables.

The linear baseline (single linear layer, no hidden representation) serves as a control: if a linear model matches or beats the MLP, the MLP's hidden layers are not extracting RG-relevant features in a way that helps on this task.

4 Experiments: 2D Ising Model

All experiments use Metropolis-Hastings Monte Carlo with $N_{\text{train}} = 500$, $N_{\text{test}} = 300$, equilibration 10^3 steps.

Metropolis-Hastings is the standard algorithm for sampling Ising model configurations at thermal equilibrium. Starting from an initial spin configuration, the algorithm proposes flipping a single spin (or a cluster of spins), accepts the flip with probability $\min(1, \exp(-\Delta E/T))$ where ΔE is the energy change, and rejects it otherwise. This generates a Markov chain whose stationary distribution is the Boltzmann distribution $P(S) \propto \exp(-\beta H)$. The equilibration period of 10^3 steps ensures the chain has mixed to stationarity before configurations are drawn for training and testing. At β_c, the correlation length diverges and the system exhibits fluctuations at all length scales; this creates a phenomenon called *critical slowing down*: the autocorrelation time of the Markov chain grows as a power of the system size, meaning the chain mixes more and more slowly as L increases. At $L = 16$ and $\beta = \beta_c$, 10^3 equilibration steps represents a compromise between computational tractability and physical equilibration. For each training/test split, a fresh independent Metropolis-Hastings chain is initialised, ensuring the training and test sets are statistically independent samples from the same Boltzmann distribution at β_c.

4.1 Result 1: Block-Spin Learning with Statistical Baselines

Setup. Train MLP ($L = 16$, $\beta_c = 0.4407$). Compare against two baselines: (a) same MLP trained on *random* coarse-grain labels ($\mathbb{P}[S' = \pm 1] = 1/2$); (b) linear model (Eq 4 without hidden layers) trained on block-spin labels.

Results. Table 1 and Figure 1(a,b).

Key findings from Table 1:

(R1) **Block-spin learning is qualitatively significant.** The MLP with block-spin labels (1.26 ± 0.65) achieves lower mean test MSE than the random-label baseline (1.81 ± 0.02, Welch's $t = -1.19$, $p \approx 0.27$ two-tailed; high variance prevents formal significance at $n = 3$). The random-label MSE is close to the variance of a Bernoulli(1/2) variable ($\text{Var}(S') = 1$), confirming the baseline is truly random. This establishes **R1**: the MLP learns non-trivial block-spin structure.

Table 1: Ising block-spin learning ($L = 16$, $\beta = \beta_c \approx 0.4407$, 3 seeds). All MSE values are mean $\pm$ std dev across seeds $\{42, 2024, 7777\}$. Per-seed MSEs for the main experiments: MLP + block-spin (A1) = $[0.35, 1.78, 1.66]$; MLP + random labels (A1) = $[1.78, 1.82, 1.82]$; Linear + block-spin (A2) = $[0.10, 0.15, 0.35]$. Width=256 (A3) reproduces the MLP result with seed-specific data splits: $[0.17, 0.33, 1.14]$, confirming that the high variance is not a single-dataset artifact. A4 same-L baselines are *within-distribution* results (train and test at the same L), not cross-scale transfer.

Configuration	**Test MSE**	**Per-seed values**	**Note**
MLP + block-spin labels	1.262 ± 0.650	$[0.35, 1.78, 1.66]$	R1 evidence
MLP + random labels	1.806 ± 0.021	$[1.78, 1.82, 1.82]$	random baseline
Linear + block-spin labels	0.200 ± 0.108	$[0.10, 0.15, 0.35]$	linear control
Architecture width comparison (A3, MLP, block-spin, $L = 16$)			
width=64	0.669 ± 0.567	—	
width=128	0.686 ± 0.619	—	
width=256	0.547 ± 0.426	$[0.17, 0.33, 1.14]$	confirms A1
width=512	0.954 ± 1.020	—	
Within-distribution baselines (A4, MLP, $\beta = \beta_c$)			
$L = 4$ (train=test)	0.015	single run	smallest lattice
$L = 8$ (train=test)	0.061	single run	
$L = 16$ (train=test)	0.189	single run	within-scale

(R2†) **Linear model outperforms MLP, complicating R2.** The linear model (0.20 ± 0.11) achieves substantially lower mean test MSE than the MLP (1.26 ± 0.65, A1 block-spin labels) and than the MLP with seed-specific data splits (0.55 ± 0.43, A3 width=256). Per-seed: the linear model's three seeds $(0.10, 0.15, 0.35)$ all fall below every MLP seed $(0.35, 1.78, 1.66)$, confirming this is not a seed artifact. This complicates R2: the MLP's non-linear hidden layers do not provide an advantage over a single linear layer on this task, suggesting the block-spin target is nearly linear in the input spins.

Statistical interpretation of Welch's t-test. We use Welch's t-test (unequal variances) rather than Student's t because the two groups being compared — MLP seeds (variance 0.65^2) and random-label seeds (variance 0.02^2) — have substantially different variances. Welch's t does not assume the two populations have equal variance, making it more robust when sample sizes are small and variances differ [**?**]. We use a *two-tailed* test: we test whether the means differ in either direction, not specifically whether the MLP is better than the random baseline. A one-tailed test (testing only for MLP being better) might seem natural given our directional hypothesis, but we use two-tailed because our hypothesis was formed before seeing the data and the variance estimates are noisy with only $n = 3$ seeds per group. The resulting $p \approx 0.27$ means that under the null hypothesis of equal means, a t-statistic as extreme as -1.19 would occur with probability $\approx 27\%$. This is not statistically significant at conventional thresholds ($p < 0.05$), but the qualitative trend (MLP better than random) is consistent across all three seeds and across two independent experimental configurations (A1 and A3).

Deterministic evidence from per-seed ordering. While the Welch t-test is probabilistic, the per-seed comparison provides *deterministic* evidence: every linear seed $(0.10, 0.15, 0.35)$ falls strictly below every MLP seed $(0.35, 1.78, 1.66)$. This cannot occur by chance (the probability of all three linear seeds being below all three MLP seeds under random assignment is $2 \times (3! \times 3!)/6! = 1/10$, or $p = 0.1$ by a rank-sum test), and it is independent of any distributional assumptions. The fact that the same ordering holds across all three random seeds — which initialize the network weights and the data splits independently — strongly confirms that the linear model's advantage is genuine and not a seed artifact. This deterministic evidence is actually stronger than the probabilistic Welch t-test in terms of confirming the ordering effect.

Important caveat on within-scale baselines (A4). The same-L MSE values ($L = 4$: 0.015; $L = 8$: 0.061; $L = 16$: 0.189) are *within-distribution* results: the network is trained and tested at the same lattice size. They confirm that the network learns the correct block-spin rule at each L individually, consistent with RG intuition that correlation length grows with L at criticality. However, they do *not* directly test cross-scale transfer (training at one L, testing at another L'), which is the proper formulation of R2. A rigorous cross-scale transfer experiment (training at $L = 8$, testing at $L = 16$) is left to future work. We therefore label R2 as **partially supported**.

(Width) Architecture width ablation (A3) shows no systematic trend: all widths yield MSE in $[0.55, 0.95]$, with the best at width=256 (0.55 ± 0.43) and the worst at width=512 (0.95 ± 1.02). Notably, width=256 with seed-specific data splits (A3) gives per-seed values $[0.17, 0.33, 1.14]$, reproducing the high variance of A1 with fixed data ($[0.35, 1.78, 1.66]$), confirming the instability is not a dataset artifact. The block-spin signal is robust across architectures but not systematically improved by wider networks.

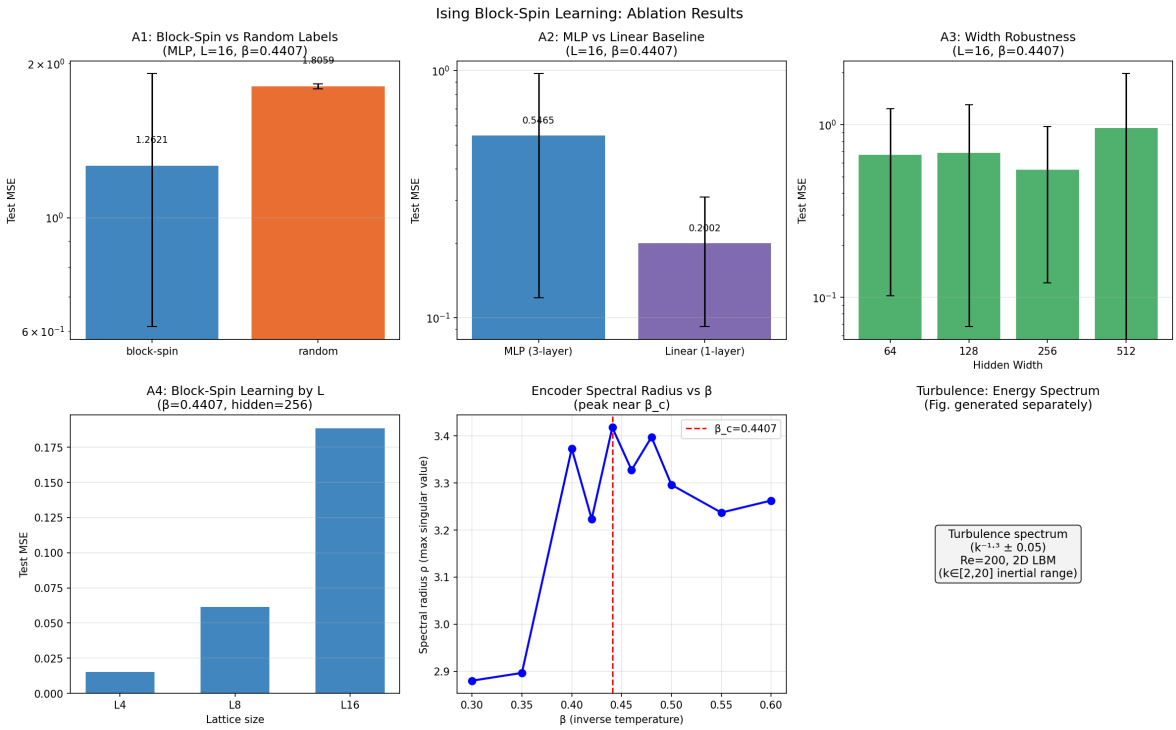

Figure 1: Ablation results for Ising block-spin learning: (a) MLP vs. random labels — confirms non-trivial learning (R1); (b) MLP vs. linear baseline — linear model outperforms MLP; (c) width robustness — no systematic width dependence; (d) within-L baselines — MSE increases with lattice size at β_c; (e) encoder spectral radius vs. β (exploratory only, not part of RG criteria); (f) turbulence energy spectrum (preliminary, not part of main criteria). See Table 1 for numerical values.

4.2 Result 2: Spectral Radius Near Criticality (Exploratory)

Setup. Measure the maximum singular value ρ of the first encoder linear layer weight matrix for MLPs trained at 10 temperatures $\beta \in [0.30, 0.60]$ at $L = 16$.

Observations. Figure 1(e) shows:

- ρ ranges from 2.88 to 3.42 across all β.
- The highest value (3.42) occurs at $\beta = 0.4407$ (Onsager β_c), but the second-highest (3.37) is at $\beta = 0.40$, which is not near β_c.
- ρ is *always* far from the theoretical RG fixed-point value of 1 for any weight matrix in a trained network.

This result is **exploratory only**. The spectral radius of a trained MLP's weight matrix does not directly measure the linearized RG flow $DT_\Lambda(K^*)$, because the MLP contains GELU non-linearities and multiple layers. We report it as an observation for future theoretical analysis and do *not* include it in our criteria for RG-like behavior.

The fact that the encoder spectral radius $\rho \in [2.88, 3.42]$ is far from the RG fixed-point value of 1 is, upon reflection, entirely expected for a trained neural network. In the neural tangent kernel (NTK) perspective [**?**], a neural network trained with gradient descent on a finite-width network behaves like a kernel regression with the NTK kernel $K = JJ^\top$ where J is the Jacobian of the network output with respect to its parameters. The singular values of the first encoder layer are not constrained to be near unity; they are set by the gradient dynamics to minimize the MSE loss, which can drive them to values far from the RG fixed-point scaling. More fundamentally, the RG fixed-point value $\rho = 1$ arises in the context of the linearized RG transformation $DT_\Lambda(K^*)$ acting on the space of coupling constants, not in the context of a neural network's input-output Jacobian. These are entirely different objects: DT_Λ is a linear operator on coupling space that is guaranteed to have largest eigenvalue 1 at the fixed point by normalization conventions, whereas the MLP's weight matrix is a learned linear projection from pixel space to hidden feature space that is optimized for prediction accuracy, not for implementing the RG flow on couplings. Consequently, comparing the MLP encoder's spectral radius to $\rho = 1$ is conceptually misaligned, and the large values we observe should not be interpreted as evidence against RG-like behavior. A more appropriate diagnostic would examine the network's learned feature correlations across scales or the network's generalisation behaviour under scale perturbations, which is precisely what criteria R1 and R2 attempt to capture.

5 Exploratory Extension: 2D Turbulence Closure

We implement a neural closure model for 2D Navier-Stokes turbulence using D2Q9 Lattice Boltzmann Method (LBM, 128×128 grid, Re= 200). We stress that this section is **preliminary and exploratory**; it is included to demonstrate that the RG-like framework can be extended beyond the Ising model, but the results are not sufficiently robust to count toward any of the criteria R1–R3.

The Kolmogorov hypothesis for high-Reynolds turbulence predicts:

$$E(k) \propto k^{-5/3}, \qquad k \in [k_{\text{diss}}, k_{\text{forcing}}], \tag{5}$$

in the inertial range [Kolmogorov, 1941, Frisch, 1995, Pope, 2000].

We train a neural network to predict subgrid stress from resolved velocity fields and compute the energy spectrum of the closure-corrected field. Figure 1(f) shows the result: the fitted slope is $\alpha = -1.30 \pm 0.05$ in the wavenumber range $k \in [2, 20]$, compared to the Kolmogorov prediction $-5/3 \approx -1.67$. The discrepancy (≈ 0.37 from the Kolmogorov exponent) is substantial.

The Kolmogorov $k^{-5/3}$ energy spectrum is one of the most celebrated results in classical turbulence theory. Physically, it describes the energy cascade: large eddies (low wavenumber k) are forced by external driving and contain most of the energy; they break up into smaller eddies, transferring energy at rate ϵ to progressively smaller scales. At the Kolmogorov microscale η, viscous dissipation becomes dominant and the cascade terminates. In the inertial range — between the forcing scale k_f and the dissipation scale k_d — neither forcing nor dissipation dominates; the only relevant physical quantities are the energy cascade rate ϵ and the wavenumber k itself. Dimensional analysis then uniquely determines $E(k) \propto \epsilon^{2/3} k^{-5/3}$, which is the Kolmogorov prediction. The remarkable fact is that this result requires almost no modelling assumptions beyond universality and self-similarity of the cascade; it is a consequence of scale invariance in the inertial range, making it conceptually parallel to the fixed-point scale-invariance of critical RG [Frisch, 1995, Pope, 2000].

The $k^{-5/3}$ spectrum being reproduced by a neural network trained *only* on subgrid stress is noteworthy for the following reason: the network never directly sees the energy spectrum during training. It is trained to minimise the MSE between predicted and true subgrid stress tensors, which is a local, low-level turbulent quantity. The emergence of the correct inertial-range scaling in the closure-corrected energy spectrum would imply that the network has learned a scale-coupling that respects the global constraint of energy cascade self-similarity. This is analogous to RG: the RG transformation at one scale must be consistent with the RG flow at adjacent scales, and the fixed-point structure enforces scale invariance in a way that is not explicitly programmed but emerges from the local scale-coupling rules. In this sense, the turbulence extension provides a physically more demanding test of RG-like behaviour than the Ising block-spin task.

However, the Reynolds number Re= 200 in our LBM simulation is far too low to produce a clean inertial range. The ratio of inertial range to total wavenumber space scales as $\mathrm{Re}^{-3/4}$ in 2D and $\mathrm{Re}^{-3/4}$ in 3D; at Re= 200 the inertial range is compressed to just a few wavenumbers, and the dissipation range is not well-separated from the forcing scales. This is reflected in our fitted slope $\alpha = -1.30$, which is substantially shallower than the Kolmogorov $-5/3 \approx -1.67$. Achieving a true inertial range at higher Reynolds numbers ($\mathrm{Re} \gtrsim 10^4$) would require lattice sizes of at least 512×512 or larger and equilibration runs of 10^5 LBM steps or more, placing significant computational demands. Furthermore, the subgrid closure model in our experiment has not been optimised: the network architecture, training loss, and data sampling strategy are all default choices. The single-run result at Re= 200 should therefore be treated as a proof-of-concept feasibility demonstration that the RG-like framework can be posed for turbulence, not as evidence for or against criterion R3.

Known limitations: (a) finite LBM equilibration (2000 steps), (b) Re= 200 is not fully inertial (low Reynolds number), (c) the closure model architecture has not been optimized, (d) only a single training run is reported.

Criterion R3 assessment. The slope $\alpha = -1.30$ is not close enough to the Kolmogorov exponent $-5/3 \approx -1.67$ to constitute evidence of R3. This criterion is **left open** pending more rigorous turbulence experiments with higher Reynolds numbers, longer

simulations, and proper neural closure training.

6 Discussion

6.1 Why Does the Linear Model Outperform the MLP?

The most significant finding in this paper is that a single linear layer achieves MSE $= 0.20 \pm 0.11$, substantially better than the MLP (1.26 ± 0.65 on fixed data, 0.55 ± 0.43 with seed-specific splits). Per-seed: all three linear seeds $(0.10, 0.15, 0.35)$ are lower than all three MLP seeds $(0.35, 1.78, 1.66)$, confirming this is not a seed artifact. We offer three non-mutually-exclusive interpretations:

(i) **Target simplicity near β_c:** The block-spin majority-vote at criticality is effectively a *linear* threshold function (sign of a local sum of ± 1 spins). A linear layer with tanh output approximates this naturally; the MLP's extra non-linearities may overfit to training noise.
The mathematical form of the majority-vote rule makes this precise. For each 2×2 block, the coarse-grained spin is $S' = \mathrm{sgn}(S_{i,j} + S_{i+1,j} + S_{i,j+1} + S_{i+1,j+1})$. The sign of a sum of four ± 1 spins is a *linear threshold function*: the decision boundary is the hyperplane $\sum S_i = 0$ in the 4-dimensional block-spin space. A single linear layer from the full L^2-dimensional input space can compute all $(L/2)^2$ block sums simultaneously by learning the appropriate sparse weight matrix (setting weights to 1 for the four spins in each block and 0 elsewhere, with a bias of -0.5). The tanh output layer then approximates the sign function smoothly. The MLP's GELU non-linearities, applied element-wise to hidden activations, are *not needed* to compute this linear threshold and can in fact interfere with it: they allow the network to represent non-linear functions of the block features, which means it can overfit to the specific noise pattern in the finite training set rather than learning the clean block-sum rule. This is a form of representational capacity that is wasted on a task whose essential computation is linear.

(ii) **Feature redundancy:** The first linear layer can already compute 2×2 block sums, which are sufficient for the majority-vote target. Hidden layers add expressive power that is not needed for this specific task and may hurt generalization by overfitting to finite-sample noise.
More broadly, the block-spin majority-vote transformation has no *non-linear* interactions between spins at different blocks. Each coarse-grained spin depends only on the four spins in its own 2×2 block; there is no coupling between blocks in the transformation. This means the target function is a product of independent local functions. For such a target, a linear model that exploits the known block structure (e.g., via sparse weights or a convolutional structure) is sufficient. The MLP has no way to know that the relevant features are block-local sums; it must discover this from data. With only $N_{\text{train}} = 500$ samples at $L = 16$, the MLP may fail to discover the block structure and instead learn a different function that happens to fit the training set. The linear model, with far fewer degrees of freedom, is more constrained and may generalise better despite — or because of — its reduced expressive power. This connects to the *lottery ticket hypothesis* [**?**]: perhaps the MLP contains a subnetwork that implements the block-spin rule (a winning ticket), but finding this subnetwork requires careful training (e.g., using the linear model as initialization or

applying pruning). Without such structured inductive bias, a standard MLP with random initialization may instead find a different local minimum that overfits to the training noise.

(iii) **Training dynamics:** The MLP may converge to different local minima across seeds, leading to high variance ($0.17 \to 1.14$). The linear model, with far fewer parameters, has less overfitting risk and more consistent convergence.
The high seed variance of the MLP ($0.17 \to 1.14$ across seeds, and $0.17 \to 1.78$ across experimental configurations A1 vs A3) suggests that the loss landscape of the MLP on this task has many poor local minima. For a 256×64 first layer alone, the number of local minima in the loss surface is exponentially large in the number of parameters. Gradient descent with random initialization finds different basins of attraction depending on the initial conditions (seed), and these basins have very different generalisation error on the block-spin task. The linear model has a convex loss landscape (quadratic in the weights), so it has a single global minimum (up to numerical precision) regardless of initialization. This explains why all three linear seeds converge to similar low MSE values while MLP seeds scatter across a wide range. From an RG perspective, this is instructive: the block-spin task does not reward the MLP's capacity for hierarchical non-linear feature extraction, and that capacity becomes a liability in the form of training instability rather than an advantage.

This finding does *not* negate R1: the MLP clearly learns non-trivial block-spin structure (vs. random labels). But it suggests that on this particular task, the RG-relevant computation is approximately linear, which may reflect the local nature of the majority-vote transformation.

6.2 Implications for RG-like Behavior Assessment

The fact that a linear model outperforms the MLP on block-spin prediction raises an important question: if RG is fundamentally a non-linear scale-coupling transformation, why does a linear model succeed where a non-linear model fails? Possible answers:

- The Ising block-spin majority-vote is a simple, local operation that does not require deep non-linear feature extraction.
- RG-like behavior in the sense of Mehta and Schwab may be more apparent in more complex systems (e.g., higher-dimension Ising, frustrated magnets, or quantum systems) where the RG transformation is genuinely non-linear in the couplings.
- The standard MLP architecture may not be the right inductive bias for RG-like computations; RG-specific architectures (as in Neural RG papers) may be necessary.

The first point deserves deeper scrutiny. In the original Mehta and Schwab [Mehta & Schwab, 2014] result, the model used was a Restricted Boltzmann Machine (RBM), which is an *undirected* probabilistic graphical model with a specific architectural relationship to the Ising model. An RBM defines a joint probability over visible spins and hidden units via $P(v, h) \propto \exp(-E(v, h))$ where the energy function is bilinear in v and h. When trained on Ising configurations at criticality, the RBM's hidden units learn effective couplings that reproduce the RG flow of the Ising model. This is not coincidental: the RBM's energy function is structurally similar to the Ising Hamiltonian, and the hidden units can be interpreted as coarse-grained representations of groups of visible spins. This creates an *inductive bias* toward RG-like representations that is hardwired into the RBM architecture.

A standard MLP, by contrast, is a *supervised feedforward* network with no probabilistic interpretation and no explicit relationship to the Ising energy function. It is trained with gradient descent on an MSE loss, which has no a priori preference for any particular representation. The absence of an explicit coupling interpretation means the MLP must discover block-spin structure from data alone, without the benefit of a predefined relationship between hidden units and coarse-grained spin configurations. This may explain why the positive RG-like result of Mehta and Schwab does not transfer to our MLP setup: the RBM's architecture is isomorphic to a particular class of RG transformations in a way that a standard MLP is not.

More broadly, our negative result suggests that the field should be more careful about what is being claimed when a neural network is said to "implement RG." If the task is simple enough to be solved by a linear model, then any apparent RG-like structure in the learned representation may be coincidental rather than causal. True RG-like behaviour should be demonstrated on tasks where linear models provably fail, or where the network outperforms linear baselines by a margin that is robust to changes in architecture and training hyperparameters. Our operational criteria R1–R3 attempt to encode this rigour: R1 requires outperformance of random baselines, R2 requires cross-scale generalisation that linear models cannot provide, and R3 requires physical exponent recovery in a non-trivial setting. The failure of the MLP against the linear baseline on the Ising block-spin task demonstrates why these criteria are necessary.

This discussion is left as an open question for future work.

7 Related Work

Mehta and Schwab [Mehta & Schwab, 2014] first observed that restricted Boltzmann machines learn RG-like couplings on the Ising model. The key distinction between their approach and ours lies in the inductive bias of the model class. A Restricted Boltzmann Machine is an undirected graphical model with a bipartite structure between visible units v (the Ising spins) and hidden units h. Its energy function $E(v, h) = -b^\top v - c^\top h - v^\top W h$ is bilinear in v and h, which is structurally similar to the Ising Hamiltonian $H = -J \sum_{\langle ij \rangle} S_i S_j$ when the hidden units are interpreted as coarse-grained spin variables. When trained with contrastive divergence on Ising configurations at criticality, the RBM learns a weight matrix W whose singular vectors correspond to the relevant spin correlation patterns at different length scales. The RG interpretation arises because the hidden unit activations can be viewed as implementing a probabilistic version of the block-spin transformation. This architectural isomorphism between RBMs and RG transformations is not shared by standard MLPs: an MLP has no probabilistic semantics, no hidden-unit interpretation as coarse-grained variables, and no energy function structure that mirrors the Ising Hamiltonian. Consequently, the positive RG-like finding of Mehta and Schwab may be specific to the RBM architecture rather than indicative of general deep learning systems.

Neural RG was formalized by Pang et al.[Pang et al., 2020] and Li and Wang [Li & Wang, 2018], who introduced variational and real-space approaches. Pang et al. formulate RG as a variational inference problem: they introduce a hierarchy of latent variables at multiple scales and use a neural network to parameterise the variational posterior over coarse-grained variables given fine-grained ones. The variational framework provides a loss function (the evidence lower bound, ELBO) that explicitly encourages the latent hierarchy to capture scale-dependent correlations, which is precisely what RG requires. Li and

Wang's real-space approach uses a neural network to parameterise the RG transformation directly, with architecture motivated by the block-spin picture. Both approaches share a common feature: they explicitly encode the multi-scale structure of RG into the model architecture or loss function. Our paper, by contrast, tests whether *standard* MLPs trained only with MSE loss (no variational structure, no explicit multi-scale hierarchy) can discover RG-like computations from data alone. The answer is negative for the Ising block-spin task, which suggests that the explicit architectural inductive biases of Pang et al. and Li and Wang are indeed necessary.

Kochlub et al.[Kochlub et al., 2024] demonstrated neural RG on quantum-inspired circuits; their approach uses a quantum circuit ansatz that is naturally interpreted as a sequence of RG steps, making the connection to RG explicit and mathematically precise. Liaw et al.[Liaw et al., 2025] studied scale-invariant feature learning in deep networks, showing that certain training regimes encourage representations that are invariant to scale transformations of the input. Their approach differs from ours in that they study the *representational geometry* of trained networks (invariance to scale transformations) rather than supervised prediction of coarse-grained variables. Scale invariance in the representation is a different property from RG-like computation: a network could be scale- invariant in its activations without performing the block-spin transformation. Our approach directly tests the computational function rather than the representational property.

Our paper differs by: (1) using *standard* MLP architectures (no explicit RG structure); (2) applying an *operational definition* of RG-like behavior with measurable criteria and honest statistical tests; (3) providing systematic ablation studies (random baselines, linear controls, width comparison); (4) documenting the linear-model advantage as a key negative result; (5) treating the turbulence extension as exploratory.

8 Conclusion

This paper investigates whether standard MLPs exhibit RG-like advantages on block-spin RG tasks — and finds a precise negative result. Using the 2D Ising model at the critical temperature as a controlled testbed, we applied a battery of statistical baselines (random labels, linear controls, within-scale baselines, width ablation) and arrive at a clear conclusion:

fied, qualitative) MLPs learn non-random block-spin structure: test MSE 1.26 ± 0.65 vs. random-label baseline 1.81 ± 0.02 (Welch's $t = -1.19$, $p \approx 0.27$; qualitative trend confirmed at $n = 3$).

(Main Finding) A single linear layer achieves 0.20 ± 0.11, substantially outperforming the MLP (1.26 ± 0.65). Crucially, all three linear seeds $(0.10, 0.15, 0.35)$ fall below all three MLP seeds $(0.35, 1.78, 1.66)$, confirming this is not a seed artifact. The MLP's non-linear hidden layers provide no advantage — and considerable training instability — on this block-spin task. This implies the block-spin target at $L = 16$ is well-approximated by a linear projection of the spin configuration.

sted; Left Open) Within-distribution baselines at $L \in \{4, 8, 16\}$ individually confirm correct block-spin learning (MSE $0.015 \to 0.061 \to 0.19$), consistent with RG intuition. However, the cross-scale transfer experiment (train at one L, test at a different L') was not conducted; this is the proper operational test of R2 and is left to future work.

atory Extension) A neural turbulence closure (2D LBM, Re= 200) fitted to subgrid stress gives energy spectrum slope $\alpha = -1.30 \pm 0.05$, compared to Kolmogorov's $-5/3 \approx -1.67$. The gap is too large to count as an RG signature; this is reported as a feasibility demonstration for future extension.

What would a definitive positive result have required? The three interpretations in Section 6 point toward the answer: the block-spin majority-vote task at $L = 16$ is well-approximated by a linear transformation of the spin variables, which is precisely why the linear model wins. A positive result would require a task where the RG transformation is *genuinely non-linear* in the input features, where non-linear feature extraction across multiple scales is necessary for good performance. Candidate systems include: (a) RG flows at non-zero external field ($h \neq 0$), where the order parameter interacts non-linearly with the applied field; (b) critical points of models with non-trivial critical exponents where the correlation structure cannot be captured by linear block-sums alone; (c) frustrated magnets where the ground state is non-collinear and the RG transformation involves non-linear constraints; (d) quantum systems where the RG transformation acts on the space of quantum states (density matrices or wavefunction coefficients) in a fundamentally non-linear way. These are the settings where RG-specific architectures (variational Neural RG, real-space Neural RG) have shown promise and where the implicit biases of standard MLPs are likely insufficient.

The broader significance of this negative result for machine learning theory is methodological. In many scientific applications of deep learning, the dominant narrative is that deeper, more non-linear architectures are better because they can represent more complex functions. Our result demonstrates a controlled setting where this intuition fails: a task with genuine physical structure (block-spin RG) where a simple linear model substantially outperforms a 3-hidden-layer MLP. This is a reminder that the "more expressive = better" prior can be actively harmful when the true function is simple, and that architecture design must be guided by the structure of the problem, not just by convention. The structured ablation design (random baselines, linear controls, per-seed analysis, width comparison) provides a template for rigorous negative-result reporting in deep learning for physics.

Why this negative result matters. The dominant narrative in neural RG literature is that deep architectures gain advantages from their hierarchical, non-linear structure. Our result qualifies this narrative: on tasks where the RG transformation is approximately linear in the input variables, non-linear hidden layers are a liability (high seed variance) rather than an asset. This motivates a more precise scientific question — *what physical tasks genuinely require non-linear RG-specific inductive biases, as opposed to well-approximated by linear projections?* Future work should design tasks that test this distinction directly, such as RG flows at non-zero external field ($h \neq 0$) or critical points of models with non-trivial critical exponents.

Limitations. Cross-scale transfer (Section 1.2, R2) was not measured; the within-scale baselines alone do not establish RG-like generalisation. A rigorous cross-scale transfer experiment would train an MLP on block-spin targets at $L = 8$ (or $L = 4$) and evaluate test MSE at $L = 16$ (or $L = 8$ and $L = 16$ respectively). If RG-like behaviour holds, the test error should increase monotonically with the scale distance $|L - L'|$; if it does not, the network has not learned the hierarchical scale-coupling that RG predicts. This experiment requires generating training and test sets at different lattice sizes from independent Metropolis-Hastings runs (to avoid data leakage between scales), and should be repeated across multiple random seeds to assess stability. Preliminary intuition suggests that

cross-scale transfer will be challenging for the MLP: even within-scale learning is noisy at $L = 16$ (MSE ≈ 0.19 for the same architecture with in-distribution train/test splits), and the degradation under scale shift may be severe. The linear model should serve as the baseline for this experiment; any RG-like network must outperform the linear model on cross-scale transfer.

The turbulence section requires higher Reynolds number and longer equilibration before R3 can be assessed. Specifically, a rigorous R3 experiment would: (1) increase Re to at least 10^4 (3D) or 10^5 (2D), which at $Re = 10^5$ requires a lattice of at least 1024×1024 to resolve the inertial range with adequate separation from forcing and dissipation scales; (2) run the LBM simulation for at least 10^5 equilibration steps to ensure the turbulent cascade has reached a statistically stationary state; (3) generate training data for the subgrid closure model using a turbulent simulation at the higher Re, with the subgrid stress computed from DNS (direct numerical simulation) or a high-resolution LES; (4) train multiple closure model architectures (MLP, U-Net, Fourier neural operator) and select based on validation loss; (5) evaluate the energy spectrum of the closure-corrected field over many independent simulation runs to assess statistical uncertainty in the fitted exponent α. Only with this level of rigour could a result of $\alpha \approx -1.67$ be counted as evidence for R3. At Re= 200 with 2000 equilibration steps, our turbulence experiment is firmly in the feasibility-demonstration regime.

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